Over-the-Air Phase Measurements and Calibration in the Upper Mid-Band: Multipath Statistics and Spatial Averaging

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***Abstract*—Unresolved multipath perturbs the phase of a dominant propagation path when measurement bandwidth is limited. We combine statistical modeling and indoor measurements to characterize phase errors and assess spatial averaging. For a dominant component and a weaker aggregate residual with uniform relative phase independent of amplitude, we derive the phase-error distribution and establish its symmetry and zero mean. Measurements across indoor environments and upper-mid-band carriers compare reduced- and full-bandwidth responses from the same records. A normalized weak-perturbation approximation based on Rayleigh residual amplitudes yields smaller empirical cumulative-distribution discrepancies than a fitted zero-mean Gaussian baseline across the evaluated pooled bandwidth–frequency configurations. For over-the-air (OTA) array calibration, circular averaging of unit channel-ratio phasors estimates relative phase in the source direction. Spatial coverage improves agreement with the paired full-bandwidth reference even when the observation count is fixed. However, spatial correlation and finite displacement constrain the averaging benefit, and residual bias can remain despite reduced variability. At a fixed observation budget, systematic sampling at approximately equally spaced positions lowers median root-mean-square calibration error and the absolute-error tolerance required for a prescribed empirical coverage compared with pooled random sampling. These findings guide spatial measurement design for narrowband phase estimation and relative array calibration.**

***Index Terms*—Measurement uncertainty, multipath propagation, narrowband measurements, over-the-air array calibration, phase measurement, spatial averaging.**

I. Introduction

Extracting the phase of a dominant propagation path (usually the line-of-sight (LOS)) is important in wireless localization, sensing, angle of arrival (AoA) estimation, and in-situ antenna-array calibration. Inter-element phase differences associated with that path encode the arrival direction and support coherent beamforming [1]. Carrier phase provides precise propagation information for Global Positioning System (GPS) positioning with integer ambiguity resolution [2], [3], indoor ranging with fifth-generation (5G) signals [4], and multiband localization [5], [6]. Phase coherence across distributed arrays also enables joint localization, synchronization, and mapping [7]. In passive sensing, inter-antenna phase differences can instead reveal the direction of a target reflection [8]. These applications share the need to extract phase information associated with a particular propagation path from a received signal that may contain several components.

Another notable application that requires extracting the phase of a dominant propagation path is *in-situ antenna-array calibration*. Calibration characterizes the installed array response so that inter-element phase differences can be used for coherent combining and direction estimation [9]. This response depends on the antenna elements, receive chains, and mutual coupling [10], [11]. For the calibration application considered here, we focus on the relative phase response in the calibration-source direction. Unresolved multipath perturbs this response by mixing contributions arriving from other directions.

Anechoic chambers suppress unwanted reflections and support controlled array measurements [12], [13]. However, chamber calibration entails dedicated facilities and measurement time; accommodating an entire installed platform may be impractical, and removing and reinstalling an array can change the response being calibrated [14], [15].

When sufficient bandwidth is available, delay-domain processing can separate paths; LOS extraction is one demonstrated use [16]. In-situ calibration with 5G reference signals has also used known transmitter positions and super-resolution delay estimation to separate multipath [15]. Sparse multiband designs exploit a large frequency span with limited occupied spectrum for joint delay and carrier-phase estimation [17]. Such approaches require access to suitable frequency resources and measurement hardware.

Bayesian in-situ calibration estimates angular responses from received signals and noisy external direction observations, but requires a correct multipath model order and resolvable paths [14].

Other methods exploit structure in the observations without requiring wideband delay separation or an anechoic chamber. Carrier-phase multipath correction can use known geometry and correlated reflections across closely spaced antennas [18]. Temporal differencing with an adaptive interval suppresses interfering paths in passive angle estimation [8], while subarray covariance smoothing enables direction finding with coherent signals [19]. These methods use spatial or temporal diversity together with an estimation procedure suited to the particular

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measurement task.

A simpler possibility for relative phase measurement is to average observations across transmitter positions which maintain the dominant-path direction and inter-element phase relation nearly constant, while other path contributions vary. If the resulting phase errors are centered on zero and sufficiently diverse, averaging can reduce their effect without resolving individual reflections.

The practical question is how much accuracy simple averaging provides under these constraints, and how its benefit depends on where the observations are collected. To this end, carrier-phase multipath models relate the error to the reflected-to-direct amplitude ratio and relative path phase [20]; scattering models link spatial correlation to the angular distribution of incident waves [21]. Uncertainty analysis for stationary autocorrelated measurements accounts for the effect of dependence on the uncertainty of a mean [22]. These statistical results provide a basis for analyzing spatial averaging, but do not by themselves quantify its accuracy in bandwidth-limited phase measurements with unresolved multipath. Over a short trajectory, repeated observations may share a similar multipath-induced phase offset, so low variability need not imply a small phase error. What remains to be assessed is how bandwidth and spatial coverage affect the residual error, and how distributing a fixed observation budget across positions changes the accuracy of the averaged estimate.

First, in section II we provide a formal definition of the considered phase-measurement and calibration problems. The subsequent analysis addresses five questions:

1) **What is the typical magnitude of the phase error caused by multipath?** In section III, we provide an empirical insights into this question by evaluating measured phase variation across different environments, carrier frequencies, and bandwidths, identifying both stable LOS-dominated conditions and cases with stronger perturbations.
2) **How does multipath amplitude relate to phase error?** Section IV addresses this question by deriving a closed form expression for the phase-error distribution for a dominant direct or reflected component and a weaker aggregate residual. It establishes symmetry and zero mean when the relative residual phase is uniform and independent of the residual amplitude, and compares the derived distribution with measured paired phase differences and a fitted zero-mean Gaussian baseline.
3) **How effective is averaging when paths cannot be separated in delay?** Section V examines averaging over finite trajectories. It compares variability across trials with the standard error within a trial to assess the effect of spatial coverage on uncertainty.
4) **Can spatial averaging support OTA array calibration?** Section VI evaluates circular averaging of unit channel-ratio phasors for relative phase calibration. It separates residual bias from variability and uses a fixed observation budget to distinguish spatial coverage from observation count.
5) **How should a limited observation budget be allocated?** Section VII compares pooled random sampling with systematic sampling at approximately equally spaced positions. It evaluates calibration root-mean-square error (RMSE) and empirical absolute-error coverage using the same circular estimator and full-bandwidth reference.

II. Primer on dominant-path phase measurements and calibration

At carrier frequency f_c, consider a complex baseband channel with an impulse response.

$$\hat{h}(t) = \sum_{\ell=1}^{L} \alpha_\ell \delta(t - \tau_\ell), \tag{1}$$

where L is the number of paths, τ_ℓ is the delay of path ℓ, and α_ℓ is its complex coefficient, including its phase at f_c. We seek $\arg(\alpha_1)$ for the selected dominant path, which may be a direct or reflected component.

For a rectangular measurement bandwidth (BW) B and approximately frequency-independent path coefficients, the normalized response is

$$\begin{aligned} h_B(t) &= \frac{1}{B}\int_{-B/2}^{B/2} H(f) e^{\mathrm{j}2\pi f t}\, df \\ &= \sum_{\ell=1}^{L} \alpha_\ell \operatorname{sinc}\big(B(t-\tau_\ell)\big), \end{aligned} \tag{2}$$

where $H(f)$ is the Fourier transform of $\hat{h}(t)$, f is the offset from f_c, and $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$, with $\operatorname{sinc}(0) = 1$. For $L = 1$, the normalization preserves peak coefficient $h(\tau_1) = \alpha_1$. When multiplet paths are present, however, each response has main-lobe width proportional to $1/B$, so reduced bandwidth increases overlap between paths that could otherwise be separated in delay.

Thus, at the target delay, we measure $h_B(\tau_1) = \alpha_1 + r_B$,[1] where r_B is the combined contribution of the remaining paths. We define path one to be *dominant* when the amplitude of the *aggregate* residual at the extraction delay is smaller than the amplitude of the component of interest, i.e., $|\alpha_1| > |r_B|$; LOS is not required. Noise, when present, is included in the same residual.

Figure 1 illustrates bandwidth-induced overlap for a synthetic example with a dominant LOS path and three non-line-of-sight (NLOS) paths. Its discrete subcarrier masking and inverse fast Fourier transform (IFFT) follow Sec. III-A; zero padding refines the delay grid without improving path resolution.

The same path overlap affects inter-element phase differences used for array calibration. A K-element array has a narrowband response.

$$h^{(k)} = \sum_{\ell=1}^{L} g_\ell a^{(k)}(\phi_\ell), \tag{3}$$

[1]The phase-error model neglects shifts of the magnitude peak and evaluates the response at τ_1. The measurements in Sec. III-A instead extract coefficients at the peak of the symbol-averaged channel impulse response (CIR) magnitude.

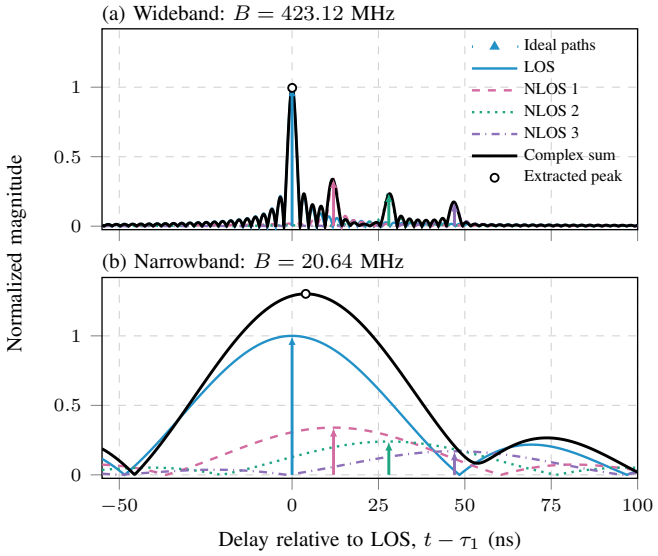


Fig. 1. Synthetic example with a dominant LOS path: (a) wideband; (b) narrowband. Arrows mark ideal paths; curves show individual band-limited magnitudes and the magnitude of their complex sum (black). Circles mark extracted peaks. Magnitudes are normalized to the isolated dominant-path peak at each bandwidth.

where g_ℓ and ϕ_ℓ are the complex incident-path gain and AoA, and $a^{(k)}(\phi)$ is the response of element k to a plane wave from direction ϕ. The vector $\boldsymbol{a}(\phi)$ is the *array signature*, incorporating geometry, antenna characteristics, and receive-chain responses [9]. An isolated dominant path gives $h^{(1)}/h^{(2)} = a^{(1)}(\phi_1)/a^{(2)}(\phi_1)$ for nonzero responses. The ratio's phase is the relative phase response sought by calibration in the source direction. Unresolved multipath mixes responses from different directions in (3), so the measured phase difference can differ from that of the dominant path. Section VI evaluates circular averaging of unit channel-ratio phasors while keeping the dominant-path direction nearly fixed. A hardware-only correction additionally requires independent knowledge of the geometric phase.

III. Experimental Observations

This section describes the measurement setup and processing procedure for OTA phase measurements targeting the LOS component. We then present an initial examination of phase errors across transmitter positions, bandwidths, carrier frequencies, and environments. These observations build intuition for the phase-error model and spatial-averaging analysis developed in the following sections.

A. Measurement setup overview

Let us first introduce the measurement setup, depicted in Fig. 2. Here, a ZCU111 radio-frequency system-on-chip (RFSoC) generates an orthogonal frequency-division multiplexing (OFDM) frame with a subcarrier spacing (SCS) of 430 kHz and 1024 subcarriers, of which 984 are active and the rest are used as guard bands. This provides a total bandwidth of 984×430 kHz $\approx$ 423 MHz. The generated signal is numerically upconverted to 1 GHz intermediate frequency (IF) and fed into the pi-radio to be upconvertd to a carrier frequency

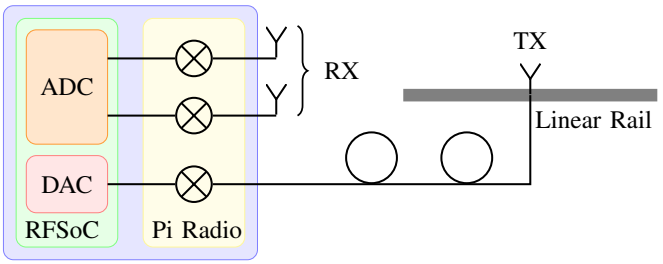
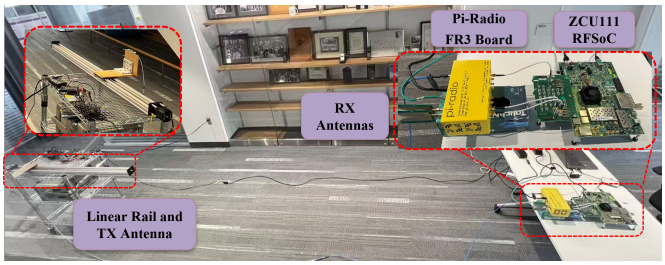


Fig. 2. Measurement setup with one TX antenna on a linear rail and two fixed RX antennas. The RFSoC provides DAC and ADC functionality, while Pi-Radio provides coherent frequency conversion. Moving the transmit antenna allows the phase difference between the receive channels to be measured along the rail.

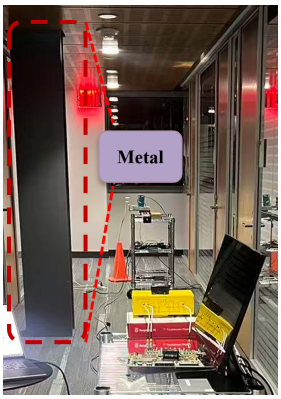
f_c in the frequency range 3 (FR3) band. In this experiment, we utilized $f_c \in \{8.75, 10.25, 12.975, 13.9, 14.9\}$ GHz[2]. The upconverted signal is fed through a long cable to a vivaldi antenna positioned on a linear rail, which can move it twoards the receiver (RX). At the RX side, we use two vivaldi antennas spaced by xxxcm which are coherently downconverted to the 1 GHz IF by the pi-radio and received by the RFSoC, which performs numerical downconversion to baseband and returns the samples through 1 Gbps Ethernet for processing. The rail is oriented in the radial direction from the RX, such that as the transmitter (TX) moves, its angle with the RX antenna positions does not change. With this setup, we sample 300 transmitter positions separated by 2 mm. At each position and carrier frequency f_c, we collect 15 OFDM frames, each containing five data symbols. Thus, each position provides $15 \times 5 = 75$ symbol-level observations per receive channel. Throughout the measurement analysis, $i \in \{1, \ldots, 300\}$ indexes position and $j \in \{1, \ldots, 75\}$ enumerates these observations, with the five symbols of each frame listed consecutively. Equivalently, frame $a \in \{1, \ldots, 15\}$ and symbol $b \in \{1, \ldots, 5\}$ map to $j = 5(a-1)+b$. This combined index retains every observation; it does not average over frames or symbols. Digital conversion uses digital-to-analog converter (DAC) and analog-to-digital converter (ADC) blocks in the RFSoC. Timing synchronization is based on correlation of repeated samples [23], using the cyclic prefix in our implementation [24]. We then demodulate the frames and divide by the known transmitted symbols to estimate the channel frequency response (CFR) of each receive channel. These responses are stored for subsequent processing. Representative measurement environments are shown in Fig. 3.

Prior to each experiment, we calibrate the response of the hardware by connecting the TX antenna port directly to each of the receivers ports (using a 20 dB [citation to the datasheet] attenuator to protect the sensitive radio-frequency (RF) input). In this configuration, we collect 200 CFR estimates. As the synchronization is not perfect, for each CFR we remove the residual delay by fitting and compensating a phase component that is linear with frequency. Subsequently, we average all CFRs to obtain an estimate of the end-to-end response of the hardware. We then divide each experimental measurement by the obtained CFR.

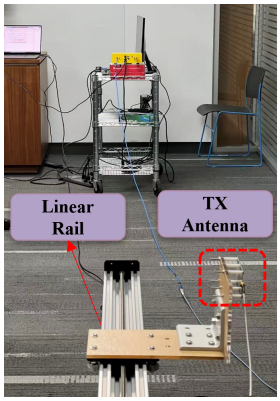
[2]The choice of frequencies is primarily due to license availability



(a)



(b)



(c)

Fig. 3. Measurement setup in different environments. (a) Hallway measurement setup. (b) Office corridor environment. (c) Conference room environment.

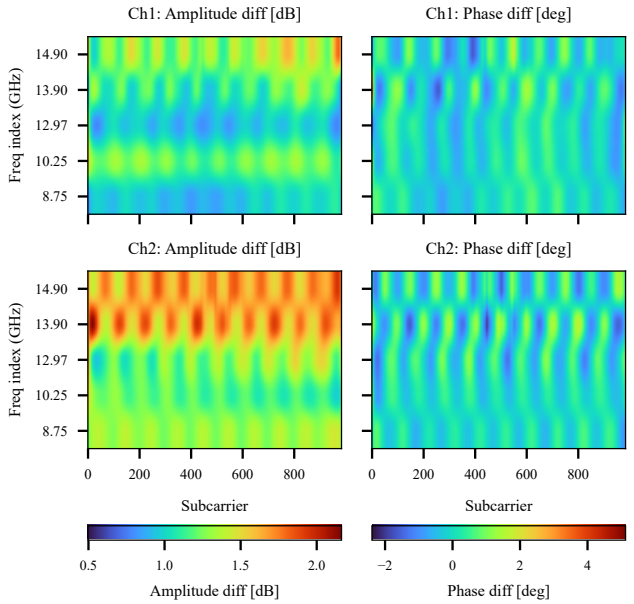


Fig. 4. Calibration response changes from before to after a representative measurement. The left column shows amplitude ratios in decibels and the right column shows wrapped phase differences in degrees. The top and bottom rows correspond to receive channels 1 and 2, respectively. The horizontal and vertical axes indicate subcarrier index and carrier frequency.

To assess changes in the hardware response during an experiment, an additional calibration measurement is collected for both receive channels immediately after the experiment. Figure 4 compares the two calibration snapshots for a representative measurement. Let C_{before} and C_{after} denote the calibration coefficients for the same receive channel, carrier frequency, and subcarrier before and after the measurement. We compute the amplitude change in decibels and the wrapped phase change in degrees as

$$\Delta A = 20 \log_{10} \left| \frac{C_{\text{after}}}{C_{\text{before}}} \right|,$$
$$\Delta \phi = \frac{180}{\pi} \arg(C_{\text{after}} C^*_{\text{before}}) .$$

The heatmaps resolve these changes across the occupied subcarriers and the five carrier frequencies for each receive channel. Zero in either map indicates no change in the corresponding quantity; positive amplitude values indicate a larger response after the measurement. In this example, the maximum absolute amplitude and phase changes are approximately 2.16 dB and 5.13°, respectively. This comparison quantifies the variation of the calibration responses over the measurement interval and provides a check on the stability of the two receive chains.

As the rail is placed in the far field of the receive array and oriented to maintain a nearly constant arrival angle, an isolated LOS component would produce a nearly constant phase difference between the two receive antennas along the trajectory. The full-bandwidth measurements provide a reference for assessing this behavior, as shown in Fig. 7. To study the impact of unresolved multipath, we label reduced-bandwidth responses as narrowband (NB) and full-bandwidth responses as wideband (WB). We compare both responses at receiver 1 while retaining the full bandwidth at receiver 2 as a common reference.

The stored channel frequency responses are first processed jointly to align the two channel impulse responses and suppress leakage. Each response is then divided by the calibration response of its receive channel. Reduced-bandwidth responses are constructed offline at receiver 1 by zeroing subcarriers at the band edges; the full-bandwidth responses of both receivers are retained. For each of these three responses, we apply 64-fold frequency-domain zero padding and an IFFT to obtain an interpolated CIR. We select the delay at which the impulse-response magnitude, averaged over the five data symbols, is largest, and extract the complex coefficient at that delay for each symbol.

We write $h^r_{B,i,j}$ for the extracted coefficient at position i and observation j, where the superscript $r \in \{1, 2\}$ identifies the receive channel and $B \in \{\text{NB}, \text{WB}\}$ denotes the selected reduced bandwidth or the full bandwidth, respectively. The run, carrier frequency, and chosen NB bandwidth are implicit; we omit i, j when they are not needed. The three coefficients above are therefore h^2_{WB} for the reference channel, h^1_{WB} for the full-bandwidth measurement, and h^1_{NB} for the reduced-bandwidth measurement. We also retain the two complex ratios $h^1_{\text{WB}}/h^2_{\text{WB}}$ and $h^1_{\text{NB}}/h^2_{\text{WB}}$. Their phases are the corresponding measurement phases relative to the common reference. These five quantities form the processed dataset. Figure 5 summarizes the complete chain from signal acquisition to these outputs. Section VI extends the same processing to obtain h^2_{NB}.

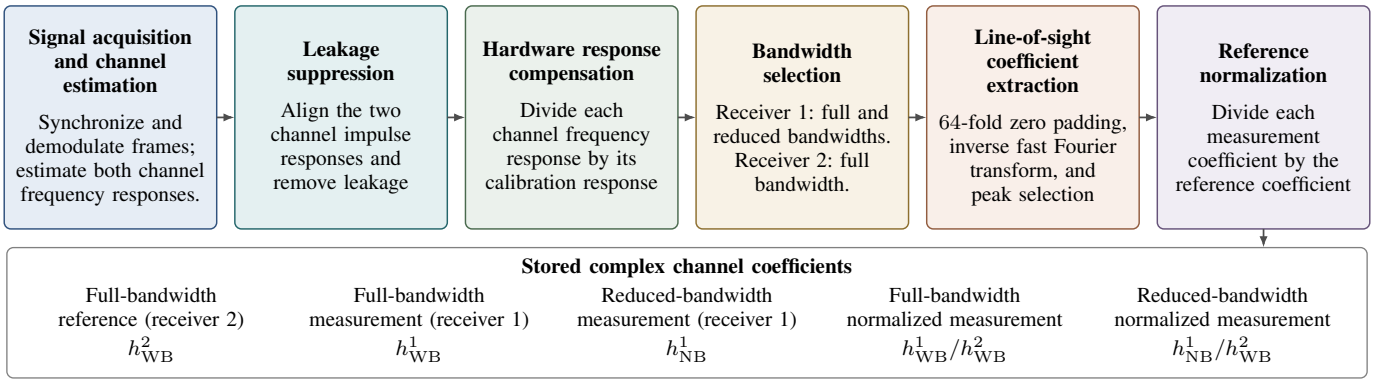


Fig. 5. Summary of the measurement and preprocessing chain. The full-bandwidth coefficient from receiver 2 serves as the reference for normalizing the full- and reduced-bandwidth coefficients from receiver 1.

TABLE I
DESCRIPTIONS OF THE MEASUREMENT SCENARIOS

Meas.	Location	Notes
1	Hall	Measurement layout shown in Fig. 3(a).
2	Lab 1	An unobstructed transmitter–receiver path; a glass curtain wall on one side; no prominent metallic objects.
3–4	MilliLab[a]	An unobstructed transmitter–receiver path; a glass curtain wall on one side; no prominent metallic objects.
5–6	Meeting room 1[a]	A wall and bookcases containing books on one side.
7–8	Cubicles[a]	Surrounded by wooden desks.
12–13	Meeting room 2[a]	A glass wall on one side and conference tables and chairs on the other.
9	Front desk	A glass wall on one side and wooden reception counters on the other.
10–11	Corridor[a]	A large metallic column on one side.
14–26	Lab 2	Multiple runs obtained by varying the setup orientation and placing sheet metal on the floor and beside the setup.

[a] Two runs per location, obtained by swapping the transmitter and receiver locations.

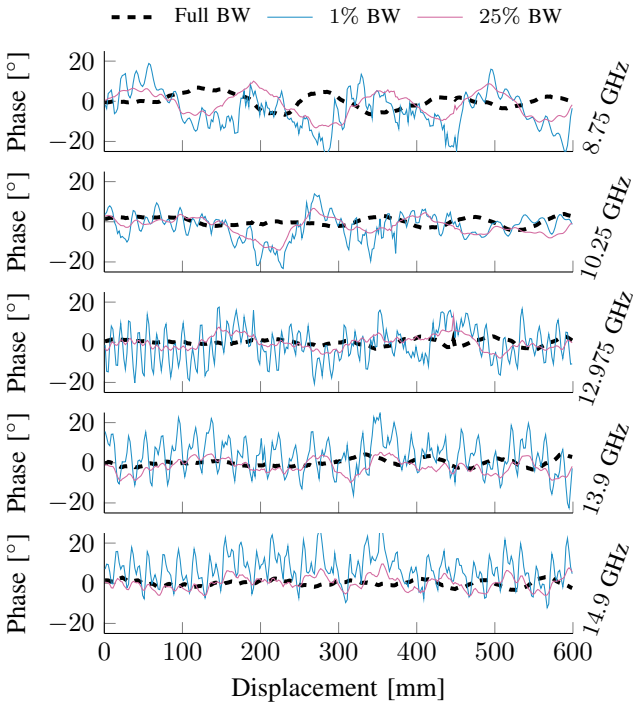


Fig. 6. Example of measured phase as a function of position for different frequencies and different bandwidths

B. Experimental observations

Using the acquisition and processing procedure described in Sec. III-A, we obtained 26 measurement runs across indoor environments, including laboratories, meeting rooms, cubicles, and hallways. Table I summarizes the locations and surrounding structures, while Fig. 3 illustrates representative measurement layouts. The runs include swapping the transmitter and receiver locations at selected sites, as well as changes in orientation and the addition of metallic reflectors in Lab 2. These configurations provide a range of multipath conditions for examining the stability of the extracted phase. Within each run, the full- and reduced-bandwidth coefficients are derived from the same acquired responses, allowing bandwidth effects to be compared under the same measurement conditions. We first examine the position-dependent phase relative to the common reference in a representative run in Fig. 6, then summarize the phase variation within each of the 26 runs in Fig. 7.

Fig. 6 shows the extracted phases at all carrier frequencies for receiver 1 with nominal bandwidths of 400 MHz, 100 MHz and 4 MHz, relative to the full-bandwidth reference at receiver 2. Here we note a few interesting properties:

1) The total phase deviation introduced by multipath is relatively small, with a peak-to-peak error in the order of 20° even at extremely low bandwidths.
2) If oberved over long enough displacement, the average value of the pahse matches reasonably well with the wideband measurements[3].
3) The deviation appear to exhibit some periodicity.

Fig. 7 shows the standard deviation of the measured phase across multiple independent measurements for three occupied bandwidth ratios (1%, 25%, 75%, and 100%). Here, we note a few interesting facts. First, in most cases the measurements

[3]Note that some deviation is expected due to the non-flat frequency response of the antenna.

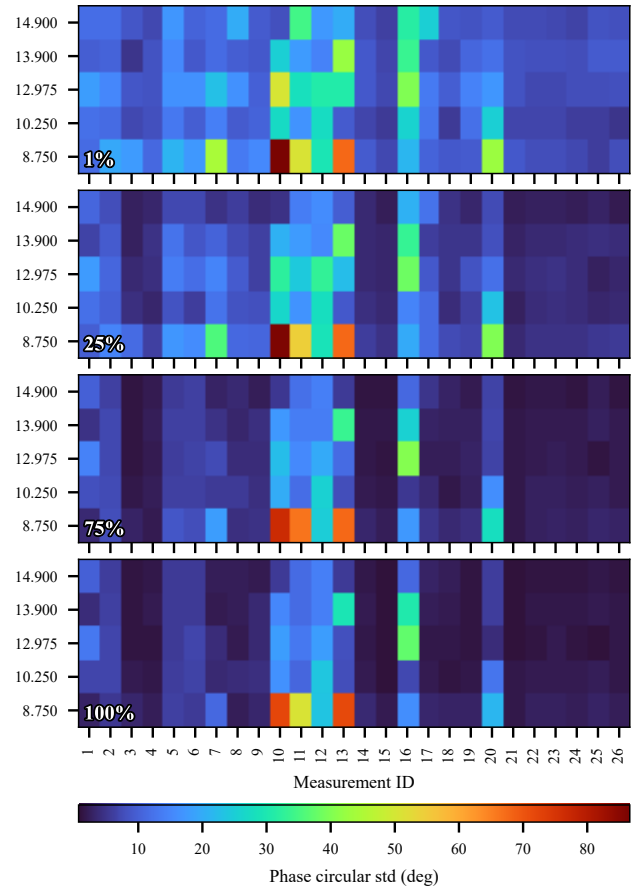


Fig. 7. Standard deviation of the measured phase across multiple independent measurements for four occupied bandwidth ratios (1%, 25%, 75%, and 100%) and different carrier frequencies.

exhibit phase variations below 20°, indicating stable LOS-dominated conditions. A few cases (e.g., measurements 10–13) show noticeably larger deviations due to strong local multipath. For example, measurements 10–11 were conducted in the narrow corridor described in Table I and shown in Fig. 3(b), where a prominent metal column between the TX and RX, together with the confined space, produced strong reflections and increased phase instability.

IV. One-shot statistics

A. Derivation of the phase distribution

To interpret the phase observations, we use the decomposition introduced in Sec. II: the measured complex coefficient consists of a dominant-path contribution and an aggregate residual containing all other paths and noise. The dominant component may be direct or reflected; the experiments target the LOS path. Dividing by its complex coefficient gives a unit-amplitude, zero-phase reference, so the normalized observation's phase is the phase error. Let β and ϕ denote the normalized residual amplitude and relative phase. To derive the phase cumulative distribution function (CDF), write the resulting coefficient as

$$h_{\text{tot}} = 1 + \beta e^{j\phi}. \tag{4}$$

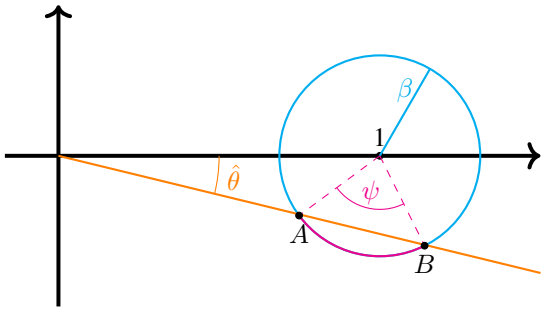


Fig. 8. Geometry of the phase CDF for $h_{\text{tot}} = 1 + \beta e^{j\phi}$, assuming a fixed residual amplitude $0 < \beta < 1$ and ϕ uniformly distributed on $(-\pi, \pi]$. The residual amplitude is smaller than the dominant-path amplitude, normalized to one, so the circle lies entirely in the right half-plane and $|\theta| \le \sin^{-1}(\beta)$.

We first analyze the regime in which the magnitude of the complex sum of all other paths and noise is smaller than the amplitude of the selected dominant path, giving $\beta < 1$ after normalization. For a fixed $0 < \beta < 1$, we assume that ϕ is uniformly distributed on $(-\pi, \pi]$. When β is random, we assume that ϕ is independent of β, so that the phase remains uniform for each given amplitude. Under $\beta < 1$, the circle in Fig. 8 lies entirely in the right half-plane, and the tangent rays from the origin give the phase support $|\theta| \le \sin^{-1}(\beta)$. The probability that the measured phase is below a fixed threshold $\hat{\theta}$ is the selected red arc length divided by the total circumference, namely $\psi/(2\pi)$. For a threshold within the support, define the signed distance

$$d = -\sin(\hat{\theta}). \tag{5}$$

The geometric distance is $|d|$. Using d with its sign accounts for both negative and positive thresholds and gives the selected arc angle

$$\psi = 2\cos^{-1}\left(\frac{d}{\beta}\right) = 2\cos^{-1}\left(\frac{-\sin(\hat{\theta})}{\beta}\right). \tag{6}$$

Thus, for $0 < \beta < 1$, the exact conditional CDF is

$$\begin{aligned} F_\theta(\hat{\theta}|\beta) &= \Pr\left[\theta \le \hat{\theta}|\beta\right] \\ &= \begin{cases} 1 & \text{if } \hat{\theta} \ge \sin^{-1}(\beta) \\ 0 & \text{if } \hat{\theta} \le -\sin^{-1}(\beta) \\ \frac{1}{\pi}\cos^{-1}\left(\frac{-\sin(\hat{\theta})}{\beta}\right) & \text{otherwise} \end{cases} \end{aligned} \tag{7}$$

and differentiation gives the exact conditional probability density function (PDF)

$$f_\theta(\theta|\beta) = \begin{cases} \frac{\cos(\theta)}{\pi\beta\sqrt{1-\frac{\sin^2(\theta)}{\beta^2}}} & \text{if } -\sin^{-1}(\beta) < \theta < \sin^{-1}(\beta) \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

At $\beta = 0$, the phase error is identically zero. For any amplitude density f_β supported on $0 \le \beta < 1$, the unconditional phase density is exactly

$$f_\theta(\theta) = \int_0^1 f_\beta(\beta) f_\theta(\theta|\beta)\, d\beta. \tag{9}$$

To model fluctuations in the residual amplitude, consider a Rayleigh distribution with scale ζ, for which

$$f_\beta(\beta) = \frac{\beta}{\zeta^2} e^{-\beta^2/(2\zeta^2)}, \quad \beta \geq 0, \qquad \Pr(\beta \geq 1) = e^{-1/(2\zeta^2)}.$$

We use this model in the weak-perturbation regime $e^{-1/(2\zeta^2)} \ll 1$, approximating the phase density in (9) by neglecting the $\beta \geq 1$ tail. For $|\theta| < \pi/2$, the support condition $|\theta| < \sin^{-1}(\beta)$ in (8) is equivalent to $\beta > |\sin\theta|$. Smaller residual amplitudes cannot produce the phase θ, so $f_\theta(\theta|\beta) = 0$ below this threshold. Hence the integral starts at $|\sin\theta|$, giving

$$f_\theta(\theta) \approx \int_{|\sin\theta|}^1 \frac{\beta e^{-\beta^2/(2\zeta^2)}}{\zeta^2} \frac{\cos\theta}{\pi\beta\sqrt{1-\sin^2\theta/\beta^2}}\, d\beta,$$

$$f_\theta(\theta) \approx \frac{\cos\theta}{\pi\zeta^2} \int_{|\sin\theta|}^1 \frac{\beta e^{-\beta^2/(2\zeta^2)}}{\sqrt{\beta^2-\sin^2\theta}}\, d\beta. \tag{10}$$

For small ζ, the exponential decay suppresses the added tail beyond $\beta = 1$, motivating a further approximation that extends the upper limit to ∞. Substituting

$$u = \sqrt{\beta^2 - \sin^2\theta}, \qquad \frac{\beta\, d\beta}{\sqrt{\beta^2-\sin^2\theta}} = du$$

into (10) with the extended upper limit, we define

$$\begin{aligned} g(\theta;\zeta) &= \frac{\cos\theta}{\pi\zeta^2} e^{-\sin^2\theta/(2\zeta^2)} \underbrace{\int_0^\infty e^{-u^2/(2\zeta^2)}\, du}_{\zeta\sqrt{\pi/2}} \\ &= \frac{\cos\theta}{\sqrt{2\pi}\zeta} e^{-\sin^2\theta/(2\zeta^2)}, \quad |\theta| < \tfrac{\pi}{2}, \end{aligned} \tag{11}$$

where g is set to zero outside this interval. The equalities in (11) evaluate the defining integral of g exactly; its use for the physical phase density is the approximation $f_\theta(\theta) \approx g(\theta;\zeta)$. To obtain a normalized model for the distribution comparison, let

$$C_\zeta = \int_{-\pi/2}^{\pi/2} g(\theta;\zeta)\, d\theta = 2\Phi(1/\zeta) - 1,$$

where Φ denotes the standard normal CDF, and define

$$\widetilde{f}_\theta(\theta;\zeta) = \frac{g(\theta;\zeta)}{C_\zeta}. \tag{12}$$

Here $C_\zeta \approx 1$ in the assumed regime. This normalization preserves the shape of (11) and gives a probability density with unit total mass. Integrating (12) from $-\pi/2$ to θ within the support gives the corresponding CDF:

$$\widetilde{F}_\theta(\theta;\zeta) = \begin{cases} 0, & \theta \leq -\pi/2, \\ \dfrac{\Phi(\sin\theta/\zeta) - \Phi(-1/\zeta)}{C_\zeta}, & |\theta| < \pi/2, \\ 1, & \theta \geq \pi/2. \end{cases} \tag{13}$$

In the weak-perturbation regime, its interior expression reduces to $\widetilde{F}_\theta(\theta;\zeta) \approx \Phi(\sin\theta/\zeta)$.

B. Properties of the phase distribution

Let us now consider the average phase error. For the amplitude model supported on $0 \leq \beta < 1$, taking the expectation of (9) over θ gives

$$\begin{aligned} E[\theta] &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^1 \theta f_\beta(\beta) f_\theta(\theta|\beta)\, d\beta\, d\theta \\ &= \int_0^1 f_\beta(\beta) \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \theta f_\theta(\theta|\beta)\, d\theta\, d\beta = 0. \end{aligned} \tag{14}$$

Here, $f_\theta(\theta|\beta)$ is even, so $\theta f_\theta(\theta|\beta)$ is odd. Under the stated uniform-phase and independence assumptions, the phase error therefore has exactly zero mean for any amplitude distribution supported on $0 \leq \beta < 1$. The normalized Rayleigh approximation $\widetilde{f}_\theta$ is also even and has exactly zero mean. Thus, for independent and identically distributed (i.i.d.) phase-error measurements satisfying these assumptions, the sample mean converges to zero.

C. Experimental validation

We assess how closely the normalized weak-perturbation model in (12) describes the measured phase-error distribution in our LOS setting, using a zero-mean Normal distribution as a baseline.

The experimental observable is the phase difference between the reduced- and full-bandwidth coefficients from receiver 1,

$$\Delta_n = \mathrm{Arg}\left(h^1_{\mathrm{NB},n}\, \overline{h^1_{\mathrm{WB},n}}\right). \tag{15}$$

The overline denotes complex conjugation, and the phase is wrapped to $[-\pi, \pi)$. If $\theta^{(1)}_{\mathrm{NB},n}$ and $\theta^{(1)}_{\mathrm{WB},n}$ denote the corresponding errors relative to the true LOS phase, then

$$\Delta_n = \mathrm{wrap}_{[-\pi,\pi)}\left(\theta^{(1)}_{\mathrm{NB},n} - \theta^{(1)}_{\mathrm{WB},n}\right). \tag{16}$$

Only when the residual full-bandwidth phase error is sufficiently small can we use $\Delta_n \approx \theta^{(1)}_{\mathrm{NB},n}$ as a proxy for the theoretical dominant-path phase error, with LOS as the target in these measurements. Both coefficients are derived from the same acquired responses, so their errors may be correlated; independence between them is not assumed. In the empirical fits below, we write $\theta_n := \Delta_n$ for the measured samples. These fits assess the model's description of the paired phase differences, subject to this reference approximation.

Table II first checks how often observations fall outside the approximate model's support. Table III then reports the estimated scale parameters used to construct the theoretical curves. Figure 9 compares the empirical and fitted CDFs, and Table IV quantifies their agreement using the empirical cumulative distribution function (ECDF) mean-square discrepancy defined below. Each bandwidth–frequency configuration pools observations from all 26 experiments, giving 30 configurations across six bandwidth ratios and five carrier frequencies. Each configuration contains $N = 26 \times 300 \times 75 = 585{,}000$ scalar phase observations: 26 experiments, 300 positions per experiment, and all 75 observations at each position. The index $n \in \{1, \ldots, N\}$ pools the run index and the (i, j) observation indices defined in Sec. III-A. Spatial and temporal correlation, as well as differences between measurement environments,

TABLE II
PERCENTAGE OF PAIRWISE PHASE-ERROR SAMPLES OUTSIDE THE APPROXIMATE MODEL'S SUPPORT ($|\theta| > \pi/2$), POOLED OVER ALL 26 EXPERIMENTS. ALL SAMPLES ARE RETAINED IN THE SUBSEQUENT FITS.

Bandwidth	Frequency (GHz)				
Percent	8.750	10.250	12.975	13.900	14.900
1%	0.304%	0.044%	0.485%	0.423%	0.584%
5%	0.206%	0.026%	0.344%	0.375%	0.022%
10%	0.209%	0.000%	0.362%	0.332%	0.000%
25%	0.241%	0.000%	0.092%	0.268%	0.000%
50%	0.357%	0.000%	0.059%	0.290%	0.000%
75%	0.463%	0.000%	0.058%	0.333%	0.000%

TABLE III
PHASE-SPREAD PARAMETER $\hat{\zeta}$ ESTIMATED FROM ALL PHASE-ERROR SAMPLES, WITHOUT PHASE-SUPPORT FILTERING, FOR EACH BANDWIDTH RATIO AND CARRIER FREQUENCY.

Bandwidth	Frequency (GHz)				
Percent	8.750	10.250	12.975	13.900	14.900
1%	0.300	0.200	0.276	0.200	0.197
5%	0.285	0.170	0.239	0.174	0.129
10%	0.274	0.163	0.225	0.162	0.116
25%	0.239	0.146	0.194	0.140	0.099
50%	0.175	0.103	0.136	0.102	0.068
75%	0.106	0.051	0.065	0.067	0.035

mean that they need not be independent or identically distributed; N is the number of recorded values, not an effective independent sample count.

The approximation assumes $\Pr(\beta \geq 1) \ll 1$, so that $\theta \in (-\pi/2, \pi/2)$ with high probability. Table II reports the percentage of samples with $|\theta| > \pi/2$ across all bandwidths and frequencies. The proportions are below 1% in every configuration, with a maximum of 0.584%. Thus, more than 99% of the observations in each configuration lie within $[-\pi/2, \pi/2]$, the support of the normalized approximation. The small out-of-support probability mass supports using this finite-support approximation to describe the bulk of the measured distribution. All observations, including those with $|\theta| > \pi/2$, are retained when estimating the parameters in Table III and comparing the CDFs in Fig. 9 and Table IV; no phase-support filtering is applied. The theoretical and Normal fits therefore use the same complete samples, and the empirical CDF retains the observed tails. With phase referenced to the dominant path, $|\theta| > \pi/2$ implies $\beta > 1$, but the converse does not hold. The measured phase-tail fraction alone therefore neither verifies the accuracy of the full-bandwidth reference nor establishes the Rayleigh amplitude assumption or a negligible value of $\Pr(\beta \geq 1)$.

Next, to estimate ζ from measured phase errors, we use the trigonometric moment of the normalized approximate model,

$$\mathbb{E}_{\widetilde{f}_\theta}[\sin^2(\theta)] = \int_{-\pi/2}^{\pi/2} \sin^2(\theta)\, \widetilde{f}_\theta(\theta;\zeta)\, d\theta. \tag{17}$$

Substituting

$$u = \frac{\sin(\theta)}{\sqrt{2}\,\zeta}, \qquad \cos(\theta)\, d\theta = \sqrt{2}\,\zeta\, du,$$

yields $\sin^2\theta = 2\zeta^2 u^2$ and the exact relation within this normalized model

$$\mathbb{E}_{\widetilde{f}_\theta}[\sin^2(\theta)] = \frac{2\zeta^2}{C_\zeta\sqrt{\pi}} \int_{-1/(\sqrt{2}\zeta)}^{1/(\sqrt{2}\zeta)} u^2 e^{-u^2}\, du. \tag{18}$$

In the assumed weak-perturbation regime, $C_\zeta \approx 1$ and the limits can be extended to $\pm\infty$, giving

$$\mathbb{E}_{\widetilde{f}_\theta}[\sin^2(\theta)] \approx 2\zeta^2 \int_{-\infty}^{\infty} \frac{u^2}{\sqrt{\pi}} e^{-u^2}\, du. \tag{19}$$

Using the Gaussian moment identity

$$\int_{-\infty}^{\infty} u^2 e^{-u^2}\, du = \frac{\sqrt{\pi}}{2},$$

we obtain the leading-order moment relation

$$\mathbb{E}_{\widetilde{f}_\theta}[\sin^2(\theta)] \approx \zeta^2. \tag{20}$$

Matching this approximate moment to the sample moment, we retain the closed-form estimator below, using all pooled measured values $\theta_n = \Delta_n$, including those outside the model support. The equality defines the estimator; its moment-based justification uses the approximation above.

$$\hat{\zeta} = \sqrt{\frac{1}{N}\sum_{n=1}^{N} \sin^2(\theta_n)}. \tag{21}$$

The largest estimate in Table III, before rounding, is $\hat{\zeta} \approx 0.3002$. Under the assumed Rayleigh amplitude model, this gives

$$\Pr(\beta \geq 1) \approx 0.389\%, \qquad 1 - C_{\hat{\zeta}} \approx 0.0865\%.$$

These values quantify the amplitude-tail probability and the normalization correction under the model, respectively. In particular, the former is not a directly measured amplitude-tail probability and does not validate the Rayleigh assumption.

Table III shows that $\hat{\zeta}$ decreases with increasing occupied bandwidth at each carrier frequency. This observation is consistent with improved separation of the LOS component from unresolved multipath at wider bandwidths. A smaller $\hat{\zeta}$ means a smaller empirical $\sin^2\theta$ moment and, in the weak-phase regime, a smaller phase spread.

The frequency dependence is not monotonic. Although the estimates at 14.900 GHz are smaller than those at 8.750 GHz for every bandwidth ratio, the values at 12.975 GHz exceed those at 10.250 GHz. Frequency-dependent propagation and residual system effects may contribute to these differences, but the present measurements do not isolate their causes or establish increased reflection attenuation or decorrelation at higher frequencies.

As a baseline, we consider how a zero-mean Normal distribution with standard deviation

$$\hat{\sigma} = \sqrt{\frac{1}{N}\sum_{n=1}^{N} \theta_n^2}. \tag{22}$$

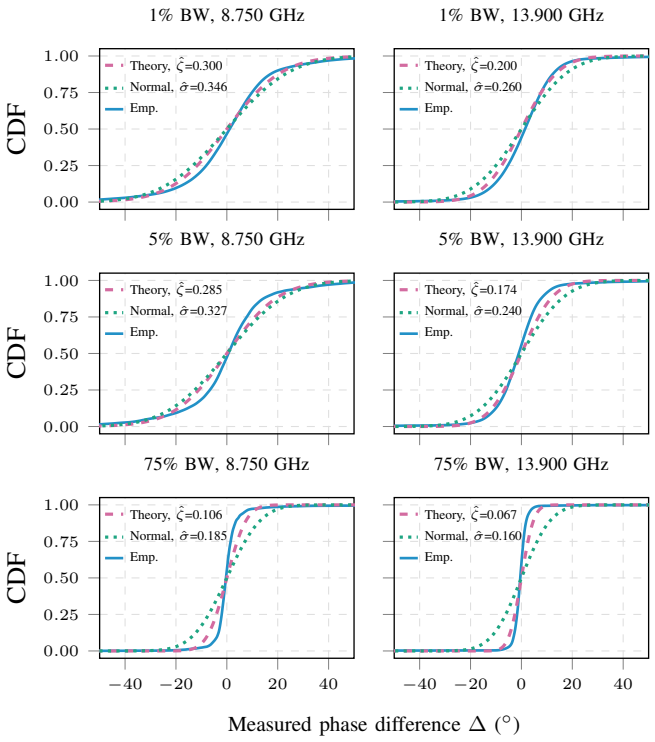


Fig. 9. Empirical, Normal, and normalized theoretical CDFs of the measured reduced-minus-full-bandwidth phase difference for bandwidth ratios of 1%, 5%, and 75% at 8.750 GHz and 13.900 GHz. The empirical CDF and both fitted models use all samples, including those outside $[-\pi/2, \pi/2]$. The theoretical curve is evaluated using (13).

fits the measured phase errors. This estimator is the root mean square (RMS) of the phase errors, with the mean fixed at zero.

Figure 9 compares this fitted Normal baseline with the normalized approximation in (12). The two scales are estimated using different moments: $\sin^2\theta$ for $\hat{\zeta}$ and θ^2 for $\hat{\sigma}$. Large phase errors affect the RMS estimate more strongly, whereas their contribution to $\sin^2\theta$ is bounded and nonmonotonic over the measured phase range. Thus, differences between the fitted CDFs reflect both the distribution shapes and the scale estimators.

To quantify goodness of fit, let $\theta_{(1)} \le \cdots \le \theta_{(N)}$ denote the sorted phase-error observations for one bandwidth–frequency configuration. For a fitted model CDF F, we compute the ECDF mean-square discrepancy

$$D_{\mathrm{ECDF}}(F) = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{i}{N} - F\big(\theta_{(i)}\big)\right]^2. \tag{23}$$

Here i/N is the empirical plotting position used for the i-th ordered observation. The same observations and plotting positions are used for both fitted CDFs: the normalized theoretical model and $F_{\mathrm{N}}(\theta) = \Phi(\theta/\hat{\sigma})$ for the Normal baseline. We evaluate the theoretical CDF directly from the analytic expression (13), both for the curves in Fig. 9 and at every ordered observation in (23). Thus, the scores do not depend on a numerical integration grid or interpolation of tabulated CDF values. The discrepancy is dimensionless, and smaller values indicate closer agreement with the empirical CDF under this criterion. This descriptive measure is a squared-CDF discrepancy of the Cramér–von Mises (CvM) type, rather than the standard one-sample CvM test statistic. Table IV reports $100D_{\mathrm{ECDF}}$; an entry of 0.64, for example, represents $D_{\mathrm{ECDF}} = 0.0064$.

TABLE IV
ECDF MEAN-SQUARE DISCREPANCY FOR THE NORMALIZED THEORETICAL MODEL (THEO) AND THE ZERO-MEAN NORMAL BASELINE (NORM). EACH ENTRY REPORTS $100D_{\mathrm{ECDF}}$, WITH D_{ECDF} DEFINED IN (23).

Percent	Frequency (GHz)									
	8.750		10.250		12.975		13.900		14.900	
Percent	Theo	Norm	Theo	Norm	Theo	Norm	Theo	Norm	Theo	Norm
1%	0.16	0.33	0.25	0.31	0.59	0.80	0.20	0.55	0.74	1.09
5%	0.32	0.52	0.10	0.16	0.31	0.62	0.36	0.99	0.46	0.57
10%	0.31	0.54	0.15	0.18	0.38	0.79	0.33	1.02	0.46	0.51
25%	0.36	0.61	0.28	0.31	0.41	0.63	0.55	1.22	0.56	0.63
50%	0.56	1.14	0.39	0.42	0.63	0.86	0.72	1.95	0.72	0.77
75%	1.34	2.99	0.63	0.64	0.60	1.52	1.40	4.10	0.90	0.91

Across the 30 pooled configurations considered here, the normalized approximation evaluated with the analytic CDF yields smaller ECDF discrepancies than the zero-mean Normal baseline fitted by RMS. Table IV reports these comparisons; some differences are small at the displayed precision. Figure 9 shows six of these configurations: bandwidth ratios of 1%, 5%, and 75% at 8.750 GHz and 13.900 GHz. This is a descriptive comparison on the same pooled samples used to estimate the model parameters; it does not establish statistical significance or agreement in every individual measurement environment.

A smaller $\hat{\zeta}$ does not necessarily imply a closer model fit. For example, at 8.750 GHz, increasing the bandwidth ratio from 1% to 75% reduces $\hat{\zeta}$ from approximately 0.300 to 0.106, while the Theo discrepancy in Table IV increases from 0.16×10^{-2} to 1.34×10^{-2}. At 75% bandwidth, the empirical CDF in Fig. 9 rises more sharply near zero than the fitted theoretical CDF, indicating a more concentrated central part of the measured distribution. The estimate in (21) uses a single empirical moment and does not minimize the ECDF discrepancy. Thus, a smaller measured $\sin^2\theta$ moment can coexist with a larger mismatch in distribution shape.

The normalized model itself has a Gaussian weak-perturbation limit. As $\zeta \to 0$, its probability mass concentrates at phase angles of order ζ, where $\sin\theta \approx \theta$ and $\cos\theta \approx 1$. Since $C_\zeta = 2\Phi(1/\zeta) - 1 \to 1$, we also have $C_\zeta \approx 1$ in this regime. Using these three approximations, (12) reduces to

$$\widetilde{f}_\theta(\theta;\zeta) \approx \frac{1}{\sqrt{2\pi}\,\zeta}\exp\left(-\frac{\theta^2}{2\zeta^2}\right), \tag{24}$$

which is the density of a zero-mean Normal distribution with variance ζ^2. This limit describes a relation between the two model families under weak perturbations. Increasing the carrier frequency does not by itself imply $\zeta \to 0$, and a small fitted $\hat{\zeta}$ does not guarantee that the measured distribution is Gaussian.

These comparisons support using the normalized approximation to describe the pooled phase-error CDFs considered here, subject to the stated weak-perturbation assumptions and fitting procedures.

V. Impact of the Spatial Correlation

Under the symmetry assumptions in Sec. IV, averaging independent phase errors reduces their variation around zero. Here we use repeated subsets of the measurements to examine how the available transmitter positions affect averaging uncertainty.

A. Sampling and uncertainty measures

For each experiment, bandwidth ratio, and carrier frequency, the quantity analyzed is the paired phase difference at receiver 1,

$$e_{i,j} = \frac{180}{\pi} \operatorname{wrap}_{[-\pi,\pi)}\left(\arg h^{1}_{\mathrm{NB},i,j} - \arg h^{1}_{\mathrm{WB},i,j}\right), \quad (25)$$

in degrees, where i indexes position and $j \in \{1,\dots,75\}$ indexes the combined frame–symbol observations defined in Sec. III-A. Thus, the full-bandwidth measurement is the reference for this comparison; it is not assumed to be an error-free LOS phase.

We retain the first $M \le 300$ positions of each recorded trajectory. For each window containing $w \le M$ consecutive positions, we choose its starting index uniformly from $\{1,\dots,M-w+1\}$ and pool its $75w$ phase differences. We draw $N_s = 50$ observations uniformly *without replacement* from this pool and repeat the procedure for $T = 1000$ trials. The evaluated window sizes are $w = 1, 6, 11, \dots$ up to M; for $M = 300$, the last evaluated size is 296. The plots use nominal lengths $D = (2\,\mathrm{mm})w$ and $D_{\max} = (2\,\mathrm{mm})M$; the first-to-last position separations are $2(w-1)$ and $2(M-1)$ mm, respectively.

For the sampled values $e^{(t)}_{w,n}$ in trial t, we compute the arithmetic mean and sample variance,

$$\bar{e}^{(t)}_{w} = \frac{1}{N_s}\sum_{n=1}^{N_s} e^{(t)}_{w,n}, \quad (26)$$

$$s^{2(t)}_{w} = \frac{1}{N_s-1}\sum_{n=1}^{N_s}\left(e^{(t)}_{w,n} - \bar{e}^{(t)}_{w}\right)^2. \quad (27)$$

The two plotted uncertainty measures are

$$\sigma_{\mathrm{inter}}(w) = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(\bar{e}^{(t)}_{w} - \bar{e}_{w}\right)^2}, \quad (28)$$

$$\sigma_{\mathrm{intra}}(w) = \frac{1}{T}\sum_{t=1}^{T}\frac{s^{(t)}_{w}}{\sqrt{N_s}}, \quad (29)$$

where $\bar{e}_w = T^{-1}\sum_t \bar{e}^{(t)}_w$. Solid curves show the standard deviation of trial means, σ_{inter}; dashed curves show the mean within-trial standard error, σ_{intra}. The latter uses $s/\sqrt{N_s}$ as an independent-sample benchmark [22], without a finite population correction.

Small windows can have similar phases internally but different local means at different positions. This yields a small within-trial standard error and a larger spread of trial means. Enlarging the window averages over more of this spatial variation. We quantify the agreement of the two uncertainty measures by

$$\Delta(w) = \left|\sigma_{\mathrm{inter}}(w) - \sigma_{\mathrm{intra}}(w)\right|. \quad (30)$$

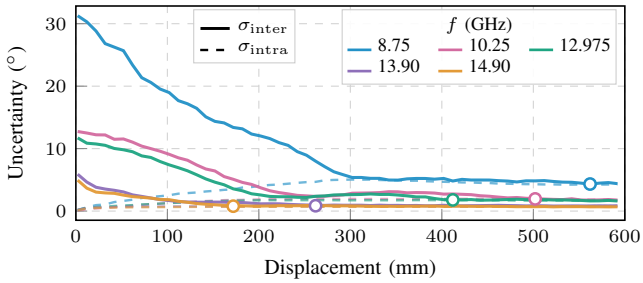


Fig. 10. Phase-averaging uncertainty for a Lab 2 measurement with a side reflector, 5% bandwidth, and 300 available positions. Solid: inter-trial standard deviation; dashed: mean within-trial standard error. Circles mark agreement within 0.23° over three successive evaluated window sizes.

B. Results and interpretation

Single-experiment example: Figure 10 uses one Lab 2 measurement with a side reflector, a 5% bandwidth ratio, and $M = 300$ positions. Across all five frequencies, the spread of the trial means decreases with window size, while the within-trial standard error initially increases as the window includes more spatial variation. The curves approach nonzero uncertainty levels that depend on the phase spread at each frequency and the fixed sample count. At 8.750 GHz the inter-trial standard deviation decreases from about 31.3° at one position to 4.4° at 296 positions; at 14.900 GHz it decreases from 4.9° to 0.7°.

The circles in this figure use a tolerance of 0.23°: we find the first three successive evaluated window sizes satisfying $\Delta(w) \le 0.23°$ and place the circle at their mean length and the mean of the two uncertainty curves over those sizes. The marked lengths decrease from 562 mm at 8.750 GHz to 172 mm at 14.900 GHz within this fixed available trajectory.

Effect of the available trajectory length: Figure 11 repeats the analysis for 20 values of M across all 26 experiments, six bandwidth ratios, and five frequencies. Shorter available trajectories are prefixes of the same recorded runs. Here, and in Fig. 12, the convergence length is the first evaluated length in a sequence of five satisfying $\Delta(w) \le 0.5°$. If no full sequence is found, the calculation accepts a shorter terminal sequence when all remaining evaluated sizes satisfy the tolerance; cases without either are excluded.

At each $D_{\max}$, Fig. 11 averages the qualifying convergence lengths across experiment, bandwidth, and frequency. The shaded band is one standard deviation across these values, not a confidence interval for the mean. The three smallest tested maximum lengths (20, 26, and 32 mm) are omitted from the plot. The dashed diagonal denotes equality with the available length, and the inset reports $100\,\overline{D}_{\mathrm{conv}}/D_{\max}$. For example, the mean convergence length increases from 126.8 mm at $D_{\max} = 200$ mm to 274.3 mm at 600 mm, while its relative length decreases from 63.4% to 45.7%. The mean length thus grows less than proportionally over this interval. This trend depends on the tolerance, finite trajectory, and exclusion of

Their agreement is an empirical stability criterion, not a proof of independent samples or an exact equality implied by the central limit theorem (CLT). It also does not establish that the mean phase error is zero.

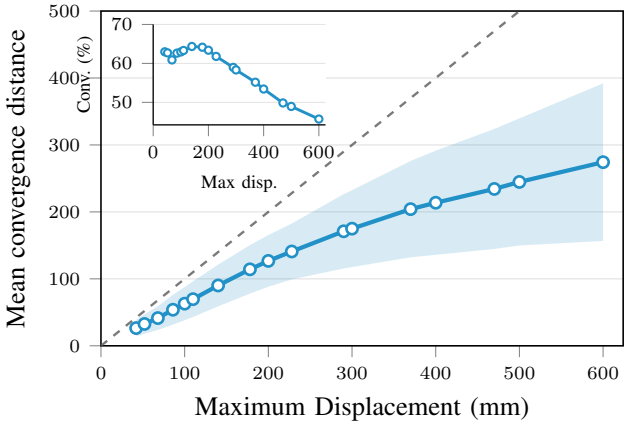


Fig. 11. Mean nominal convergence length versus available trajectory length, aggregated over qualifying configurations from 26 experiments using the 0.5° criterion. Shading shows ± one standard deviation. The diagonal denotes equal lengths; the inset shows the mean convergence length as a percentage of the available length.

nonqualifying cases.

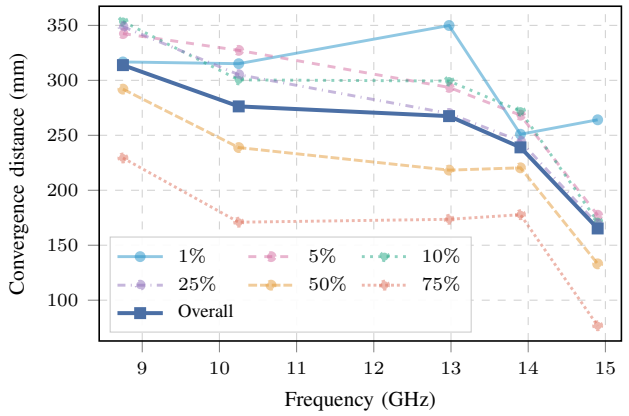


Fig. 12. Mean nominal convergence length versus frequency for the 19 experiments included in this export, with 300 available positions and the 0.5° criterion. Faded curves show means for individual bandwidth ratios; the highlighted curve averages over bandwidth ratios and included experiments.

Frequency and bandwidth dependence: Figure 12 fixes $M = 300$ and groups the qualifying convergence lengths by bandwidth ratio and carrier frequency. The current export includes 19 of the 26 experiments because the filename filter omitted seven runs; the averages therefore describe this subset. Each faded curve averages across these experiments at one bandwidth ratio, and the highlighted curve averages across all six bandwidth ratios and the included experiments at each frequency. The overall mean decreases from 313.8 mm at 8.750 GHz to 165.2 mm at 14.900 GHz. The 75% bandwidth curve lies below the narrower-bandwidth curves at all five frequencies, consistent with less spatial averaging being needed when more multipath is resolved. However, individual bandwidth curves are not uniformly monotonic in frequency; for example, the 1% curve rises between 10.250 and 12.975 GHz. These are trends in the measured subset under a fixed absolute tolerance, rather than direct estimates of a propagation correlation length.

VI. Narrowband OTA phase calibration

The previous section studies variability in averaged paired phase differences. Phase calibration estimates a persistent relative phase and applies its inverse to a receive channel. These relative antenna-path phases govern coherent combining. Broadband delay-domain measurements can reduce reflection-induced calibration errors [16]; here we assess the error from replacing a wideband phase estimate with spatial averaging at reduced bandwidth.

A. Calibration observable and identifiability

For one measurement at carrier frequency f_c, with position and observation indices i, j as in Sec. III-A, define

$$q^{\mathrm{NB}}_{i,j} = \frac{h^1_{\mathrm{NB},i,j}}{h^2_{\mathrm{NB},i,j}}, \qquad q^{\mathrm{WB}}_{i,j} = \frac{h^1_{\mathrm{WB},i,j}}{h^2_{\mathrm{WB},i,j}}. \tag{31}$$

Both NB coefficients use the same spectral mask and the same acquired (i, j) observation. The WB ratio is the fourth stored quantity in Fig. 5; the fifth quantity is the mixed-bandwidth ratio $h^1_{\mathrm{NB}}/h^2_{\mathrm{WB}}$. Here q^{NB} instead uses NB coefficients from both receivers. For $B \in \{\mathrm{NB}, \mathrm{WB}\}$, the phase $\theta^B_{i,j} = \operatorname{Arg} q^B_{i,j}$ refers to one of the 75 observations at position i. Figure 6 uses the earlier mixed-bandwidth ratio and averages complex ratios within each frame. Here all symbols have equal phase weight. Phases are in radians unless stated otherwise; Arg and wrap use $[-\pi, \pi)$.

To distinguish calibration from propagation in our LOS measurement geometry, write

$$\begin{aligned} \theta^B_{i,j} &= \operatorname{Arg} q^B_{i,j} \\ &= \operatorname{wrap}\left(\theta_{\mathrm{hw}} + \theta_{\mathrm{geom},i} + \varepsilon^B_{i,j}\right), \end{aligned} \tag{32}$$

where θ_{hw} is the residual relative receive-path phase after cable compensation, $\theta_{\mathrm{geom},i}$ is the geometric LOS phase difference, and $\varepsilon^B_{i,j}$ combines perturbations of both receive phases. This differs from the single-channel dominant-path error θ in Sec. IV. For known path lengths $r_{1,i}$ and $r_{2,i}$ from transmitter position i to receivers 1 and 2, $\theta_{\mathrm{geom},i} = -2\pi f_c(r_{1,i} - r_{2,i})/c$, with c the propagation speed. A hardware-only correction requires removing this phase. A nearly radial far-field trajectory keeps $\theta_{\mathrm{geom},i} \simeq \theta_{\mathrm{geom}}$ approximately constant. Without an independently known geometric phase, only the combined phase $\theta_0 = \operatorname{wrap}(\theta_{\mathrm{hw}} + \theta_{\mathrm{geom}})$ can be targeted.

We use the preprocessing in Sec. III-A for both receivers, with NB coefficients for relative phase calibration and WB coefficients as the evaluation benchmark. We subtract no geometric phase, so the weight targets θ_0 in the calibration-source direction, including any constant geometric phase. This does not identify the original uncorrected hardware phase or assess amplitude or mutual-coupling calibration.

B. Spatial phase estimators and correction

For w consecutive positions starting at s, define $\mathcal{W}_{s,w} = \{s, \ldots, s+w-1\}$, where $1 \le w \le 300$ and $1 \le s \le 301-w$. The *all-data estimator* uses all $75w$ observations, with $j =$

$1, \ldots, 75$ at each position in the window, normalizing each ratio for an equally weighted circular mean [25]:

$$z_{i,j}^{\mathrm{NB}} = \frac{q_{i,j}^{\mathrm{NB}}}{|q_{i,j}^{\mathrm{NB}}|}, \tag{33}$$

$$\widehat{\theta}_{s,w}^{\mathrm{NB}} = \mathrm{Arg}\left[\frac{1}{75w}\sum_{i\in\mathcal{W}_{s,w}}\sum_{j=1}^{75} z_{i,j}^{\mathrm{NB}}\right], \tag{34}$$

The *fixed-budget estimator* uses $N_s = 50$ observations for every w. In trial t, starting at $s_w^{(t)}$, draw a set $\mathcal{A}_w^{(t)}$ of N_s distinct (i,j) indices uniformly without replacement from the window's $75w$ observations across all 15 frames, and average their phasors:

$$\widehat{\theta}_{w,50}^{\mathrm{NB},(t)} = \mathrm{Arg}\left[\frac{1}{N_s}\sum_{(i,j)\in\mathcal{A}_w^{(t)}} z_{i,j}^{\mathrm{NB}}\right]. \tag{35}$$

The estimators are used separately, not in succession. For either estimate $\widehat{\theta}^{\mathrm{NB}}$, apply the phase-only correction

$$\widehat{c}_1 = e^{-\mathrm{j}\widehat{\theta}^{\mathrm{NB}}}, \qquad \widehat{c}_2 = 1. \tag{36}$$

Here $z_{i,j}^{\mathrm{NB}} = e^{\mathrm{j}\theta_{i,j}^{\mathrm{NB}}}$. With a known geometric phase, replace $z_{i,j}^{\mathrm{NB}}$ by $z_{i,j}^{\mathrm{NB}} e^{-\mathrm{j}\theta_{\mathrm{geom},i}}$ before averaging to target θ_{hw} alone. Both circular means maximize the summed cosines of the selected phase residuals and handle the $\pm\pi$ branch cut; unnormalized ratios would weight phases by amplitude. Multiplying receiver 1 by $\widehat{c}_1$ aligns its estimated phase with receiver 2 in the calibration direction. The same construction applies to other elements with known geometry and coherent responses relative to a common reference element.

If the relative phase errors are symmetric about zero and $E[\cos\varepsilon^{\mathrm{NB}}] > 0$, the population circular mean equals θ_0. These conditions are not guaranteed by the single-channel model in Sec. IV. In finite measurements, positive spatial correlation can limit the variance reduction from averaging, while a systematic phase offset need not vanish as more positions are included.

C. Reference, metric, and evaluation protocol

We use Meas. 9 at 1%, Meas. 8 at 5%, and Meas. 15 at 25% bandwidth, with IDs from Fig. 7. All five carrier frequencies are retained for each measurement; Meas. 15 also provides the fixed-budget and bandwidth studies. Both receivers use the same mask, retaining all 300 positions and 75 observations per position. For each measurement and carrier frequency, all 22,500 WB observations define a fixed benchmark:

$$\theta_\star^{\mathrm{WB}} = \mathrm{Arg}\left[\frac{1}{22500}\sum_{i=1}^{300}\sum_{j=1}^{75}\frac{q_{i,j}^{\mathrm{WB}}}{|q_{i,j}^{\mathrm{WB}}|}\right]. \tag{37}$$

This benchmark is used only for evaluation and is unchanged across NB window lengths and bandwidth ratios. There is no frame split: the NB and WB coefficients come from the same acquired frames, making this a paired comparison with potentially correlated errors. The WB phase is an empirical reference, not error-free hardware or LOS ground truth.

For the all-data results in Figs. 13(a)–(c) and 14, enumerate all $S_w = 301 - w$ window starts for each $w = 1, \ldots, 300$. The wrapped error and RMSE in degrees are

$$\delta_{s,w} = \mathrm{wrap}_{[-\pi,\pi)}(\widehat{\theta}_{s,w}^{\mathrm{NB}} - \theta_\star^{\mathrm{WB}}), \tag{38}$$

$$E_{\mathrm{cal}}(L_w) = \frac{180}{\pi}\sqrt{\frac{1}{S_w}\sum_{s=1}^{S_w}\delta_{s,w}^2}. \tag{39}$$

Here $L_w = (2\ \mathrm{mm})(w-1)$ is the first-to-last window span (0–598 mm), and $N_{\mathrm{cal}} = 75w$ is the observation count. For the same w, $L_w = D - 2$ mm, where $D = (2\ \mathrm{mm})w$ is the nominal length in Sec. V. The NB correction's error relative to the WB correction has the same magnitude as $\delta_{s,w}$. RMSE includes both the residual mean and the spread across starts; standard deviation alone would miss the mean offset.

For the *separate fixed-budget control* in Fig. 13(d), use (35) for Meas. 15 at 25% bandwidth, with $T = 1000$ trials per w and all five carrier frequencies. In trial t, draw $s_w^{(t)}$ uniformly from $\{1, \ldots, S_w\}$ and a new set $\mathcal{A}_w^{(t)}$ as above. The wrapped error and trial RMSE against the same fixed 22,500-observation WB benchmark are

$$\delta_{w,50}^{(t)} = \mathrm{wrap}\left(\widehat{\theta}_{w,50}^{\mathrm{NB},(t)} - \theta_\star^{\mathrm{WB}}\right),$$

$$E_{\mathrm{cal},50}(L_w) = \frac{180}{\pi}\sqrt{\frac{1}{T}\sum_{t=1}^{T}(\delta_{w,50}^{(t)})^2}. \tag{40}$$

The seed is 20261005. Both this control and Sec. V draw 50 observations from a pool of $75w$. Here we average unit NB-ratio phasors and compare with the fixed full-trajectory WB benchmark; that section arithmetically averages per-observation NB–WB phase differences. It fixes sample count, not independence; L_w is the available span, which may exceed the sampled span.

D. Calibration error versus displacement

Figure 13(a)–(c) shows three illustrative cases selected from the measured curves, each evaluated at all five carrier frequencies. At 8.750 GHz, from a single position to the full trajectory, RMSE decreases from 24.83° to 2.35° for Meas. 9 at 1% bandwidth, from 17.59° to 0.27° for Meas. 8 at 5%, and from 5.56° to 1.83° for Meas. 15 at 25%. Their different measurement conditions preclude attributing differences between curves solely to bandwidth.

All 15 measurement–frequency combinations have lower full-trajectory errors than single-position errors, but the improvement need not be monotonic. For Meas. 9 at 8.750 GHz, increasing L_w from 340 to 472 mm raises RMSE from 5.27° to 7.58°. Over this interval, the mean residual relative to the WB reference increases from 4.45° to 7.27°, while the standard deviation across window starts decreases from 2.82° to 2.17°. The local rise is therefore driven by an increased mean offset despite reduced variability. Greater agreement among estimates from different window starts need not imply a smaller calibration error.

Frequency-dependent residuals also remain. Meas. 8 retains 6.68° at 12.975 GHz, compared with 0.27° at 8.750 GHz.

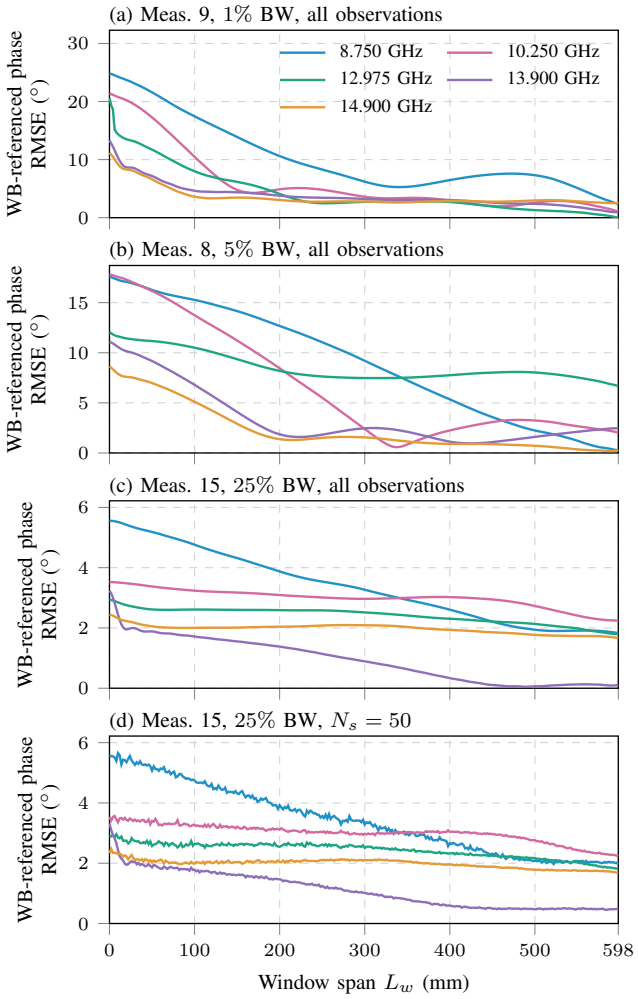


Fig. 13. WB-referenced relative phase-calibration RMSE. (a)–(c) Meas. 9, Meas. 8, and Meas. 15 at 1%, 5%, and 25% bandwidth, respectively, using $75w$ observations and all window starts. Measurement IDs follow Fig. 7. (d) Meas. 15 at 25% bandwidth: 50 observations and 1000 trials per length. All panels show five carrier frequencies with the shared legend; (c) and (d) share the same vertical scale. Both receivers use the indicated bandwidth. Each measurement and frequency uses its own full-trajectory WB benchmark. The available span is $L_w = 2(w-1)$ mm.

Spatial averaging therefore does not guarantee a small final offset at every carrier frequency.

For Meas. 15, Fig. 13(d) shows that the 8.750-GHz RMSE decreases from 5.53° to 1.96° over 0–598 mm with the count fixed at 50. The fixed-budget curves improve at all five frequencies; Table V compares both estimators. Spatial coverage therefore helps at a fixed observation count in this measurement, although the all-data residual still reaches 2.24° at 10.250 GHz.

For Fig. 14(b), let $e_{s,w} = (180/\pi)\delta_{s,w}$ be the wrapped residual in degrees. The bias (mean residual), window spread, and RMSE satisfy

$$b_{\mathrm{cal}}(w) = \frac{1}{S_w}\sum_{s=1}^{S_w} e_{s,w},$$

$$\sigma_{\mathrm{win}}(w) = \sqrt{\frac{1}{S_w}\sum_{s=1}^{S_w}\big(e_{s,w} - b_{\mathrm{cal}}(w)\big)^2},$$

$$E_{\mathrm{cal}}^2(L_w) = b_{\mathrm{cal}}^2(w) + \sigma_{\mathrm{win}}^2(w).$$

The figure plots $|b_{\mathrm{cal}}|$; window spread measures variation across the $S_w = 301 - w$ window starts, not a standard error.

TABLE V
RELATIVE PHASE-CALIBRATION ERROR FOR MEAS. 15 AT 25% BANDWIDTH. THE $N_{\mathrm{cal}} = 22{,}500$ COLUMN USES THE SINGLE FULL-TRAJECTORY WINDOW; ITS RMSE IS THAT ESTIMATE'S ABSOLUTE ERROR.

f_c (GHz)	$L_w = 0$ $N_{\mathrm{cal}} = 75$ RMSE (°)	$L_w = 598$ mm $N_{\mathrm{cal}} = 22{,}500$ Error (°)	$L_w = 598$ mm $N_s = 50$ RMSE (°)
8.750	5.56	1.83	1.96
10.250	3.52	2.24	2.28
12.975	2.94	1.80	1.82
13.900	3.24	0.11	0.48
14.900	2.45	1.67	1.68

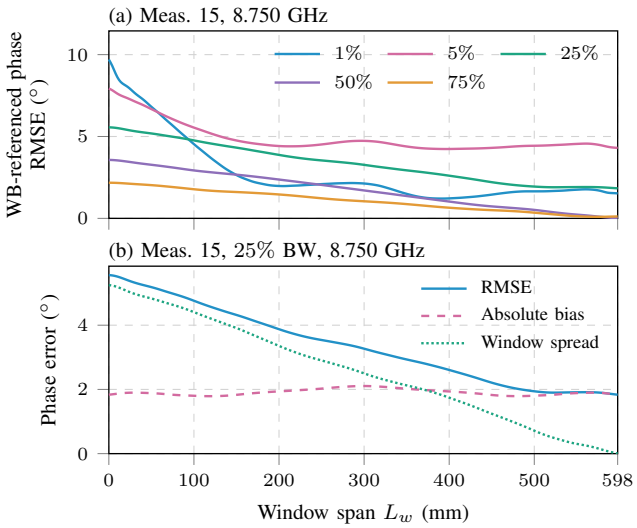


Fig. 14. Detailed calibration evaluation for Meas. 15. (a) Error versus displacement at 8.750 GHz for the five bandwidth ratios 1%, 5%, 25%, 50%, and 75%, applied to both receivers. (b) RMSE E_{cal}, absolute bias $|b_{\mathrm{cal}}|$, and window spread σ_{win} at 8.750 GHz and 25% bandwidth, matching the corresponding curve in Fig. 13(c). All-data estimates use $75w$ observations and the fixed 22,500-observation WB benchmark.

Within Meas. 15, Fig. 14(a) compares bandwidths under the same measurement conditions at 8.750 GHz. Wider bandwidth reduces the single-position error, and spatial averaging reduces the error at all five bandwidths. The absolute reduction over the full trajectory is generally larger at narrower bandwidths, where the initial error is higher. For example, increasing L_w from 0 to 598 mm reduces RMSE from 9.70° to 1.52° at 1% bandwidth, compared with 2.18° to 0.12° at 75% bandwidth. These correspond to absolute reductions of 8.18° and 2.06°, respectively. The wider-bandwidth estimate starts closer to the WB reference and has less absolute error to reduce through spatial averaging. The absolute reduction and final error are

not strictly ordered by bandwidth; the smallest full-trajectory error occurs at 50% bandwidth.

For the 25% case in Fig. 14(b), at $L_w = 500$ mm the absolute bias is $1.81°$ and the window spread is $0.72°$, giving an RMSE of $1.94°$. The mean offset already dominates before the endpoint. At 598 mm, only one window remains: RMSE equals its $1.83°$ absolute error and window spread is zero by construction. This endpoint and the overlapping windows do not establish negligible uncertainty or exact calibration. The curves use no monotonic envelope, phase-tail removal, or subtraction of the final NB offset.

VII. Sampling efficiency and reliability with limited bandwidth

Building on the calibration method in Sec. VI, we examine how to allocate a fixed observation budget across positions and how often estimates meet a given error tolerance. Results are summarized over all 26 measurements and five carrier frequencies, giving 130 measurement–frequency configurations per bandwidth. Both sampling strategies are fixed, and the comparison is descriptive; no held-out subset is used.

A. Shared estimator and observation allocation

We use the RX1-NB/RX2-NB ratio $q_{i,j}^{\mathrm{NB}}$, the two-receiver WB ratio $q_{i,j}^{\mathrm{WB}}$ in (31), and unit NB phasors $z_{i,j}^{\mathrm{NB}} = q_{i,j}^{\mathrm{NB}}/|q_{i,j}^{\mathrm{NB}}|$. The position index i and observation index $j = 1, \ldots, 75$ retain their meaning from Sec. III-A. Both receivers use the same bandwidth ratio, chosen from 1%, 5%, 25%, 50%, and 75%. The NB sampling pool contains all 75 observations at each position, giving $75w$ observations in a window. At each frequency, the WB benchmark in (37) uses all 22,500 observations across the 300 positions. There is no frame split. NB and WB share the acquired frames, so the WB benchmark is a paired empirical reference rather than independent ground truth. Both strategies use the circular estimator in (35) and the wrapped residual and RMSE in (40). Only the selected set $\mathcal{A}_w^{(t)}$ changes.

With a fixed budget of $N_s = 50$, we compare two sampling strategies:

1) *Pooled random*: as before, draw 50 distinct observations uniformly from the window's $75w$ observations.
2) *Systematic* 50×1: use 50 approximately equally spaced positions with a common random grid offset, choosing one observation uniformly per position.

For systematic sampling, the position indices are

$$i_m = s + \left\lfloor \frac{(m+U)w}{50} \right\rfloor, \quad m = 0, \ldots, 49, \quad U \sim \mathcal{U}[0,1).$$

Here s is the window's first position. A new U is drawn for each trial. Sampling is without replacement within each trial. Systematic sampling requires $w \geq 50$ to obtain 50 distinct positions; pooled sampling is feasible at every tested window length and can select several observations from the same position.

We report $L_w \in \{100, 200, 300, 400, 500, 598\}$ mm, with $L_w = 2(w-1)$ mm. Each feasible strategy uses $T = 1000$ trials per length. For paired comparisons, strategies share window starts, and each strategy shares its selected indices across frequencies and bandwidths within a run. We record the actual sampled span separately from the available span L_w. The budget counts observations, not independent samples, measurement time, or motor travel.

TABLE VI
WB phase-error tolerance ϵ_{90} (°) for 90% trial coverage, with $N_s = 50$.

BW (%)	Window span L_w (mm)					
	100	200	300	400	500	598
(a) Pooled random						
1%	23.2°	17.0°	13.2°	11.5°	11.1°	10.5°
5%	22.0°	14.4°	10.9°	9.2°	8.5°	8.1°
25%	19.9°	13.1°	9.3°	7.3°	6.2°	5.5°
50%	17.0°	11.3°	7.4°	5.9°	4.7°	3.7°
75%	13.3°	8.9°	5.8°	4.6°	3.3°	2.5°
(b) Systematic 50×1						
1%	23.1°	16.4°	12.5°	10.5°	9.7°	9.4°
5%	21.8°	13.9°	9.9°	8.5°	7.7°	7.3°
25%	19.7°	12.8°	8.4°	6.3°	5.2°	5.0°
50%	16.9°	11.0°	6.4°	5.4°	3.9°	2.8°
75%	13.2°	8.6°	5.2°	4.2°	2.6°	1.5°

B. WB agreement at a target coverage

For each strategy, bandwidth, and window span, we summarize agreement with the WB reference by the percentage of trials within an absolute phase-error tolerance. Let $\delta_{k,w,50}^{(t)}$ be the wrapped error for configuration k and trial t. Across $K = 130$ measurement–frequency configurations, define

$$\widehat{C}_{w,50}(\epsilon) = \frac{1}{KT} \sum_{k=1}^{K} \sum_{t=1}^{T} \mathbf{1}\left\{ \frac{180}{\pi} |\delta_{k,w,50}^{(t)}| \leq \epsilon \right\}, \qquad (41)$$
$$\epsilon_{90}(w) = \inf\{\epsilon \geq 0 : \widehat{C}_{w,50}(\epsilon) \geq 0.90\}.$$

Thus, ϵ_{90} is the smallest tolerance in degrees that covers at least 90% of the pooled trial errors. Each configuration contributes $T = 1000$ trials and has equal weight. These trials reuse existing records; they are not independent physical experiments. Reported tolerances are rounded upward to $0.1°$.

Table VI reports the required tolerances for window spans from 100 to 598 mm. Within each column, darker blue denotes a larger tolerance on a scale shared by both strategies. Both strategies require smaller tolerances as the window span or bandwidth increases, and systematic sampling gives a lower ϵ_{90} at every tabulated setting. Its advantage generally becomes more pronounced over longer windows: the gap is only 0.1–$0.2°$ at 100 mm, compared with 0.5–$1.1°$ at 598 mm. This is consistent with systematic sampling's more even spatial coverage under the fixed observation budget.

At 598 mm, increasing bandwidth from 1% to 75% lowers ϵ_{90} from $10.5°$ to $2.5°$ for pooled random sampling and from $9.4°$ to $1.5°$ for systematic sampling. The relative tolerance reduction from systematic sampling is about 9–11% at 1–25% bandwidth, 24% at 50%, and 40% at 75%. Thus, both methods benefit from wider bandwidth, with the largest relative advantage for systematic sampling at 75%; at 1–5%, substantial

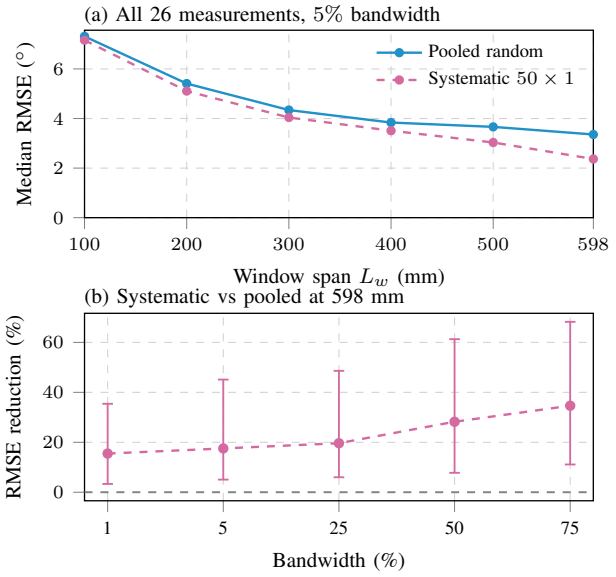


Fig. 15. Sampling comparison with $N_s = 50$ across all 26 measurements. (a) Median $E_{\mathrm{cal},50}$ across 130 measurement–frequency configurations at 5% bandwidth for pooled random and systematic 50×1 sampling over 100–598 mm windows. (b) Paired RMSE reduction of systematic relative to pooled random sampling at $L_w = 598$ mm. Points and bars show the median and interquartile range across configurations at each bandwidth; bars are not confidence intervals. Positive values indicate improvement.

TABLE VII
SAMPLING COMPARISON AT 5% BANDWIDTH AND $L_w = 598$ MM, WITH $N_s = 50$.

Sampling strategy	Median RMSE (°)	P90 RMSE (°)	ϵ_{90} (°)
Pooled random	3.36	7.95	8.1
Systematic 50×1	2.37	7.22	7.3

residual tolerances remain for both methods. At 75% bandwidth and full span, the rounded systematic tolerance covers 91.7% of the trial estimates relative to the WB reference. This aggregate coverage does not require every configuration to attain 90% individually and does not guarantee coverage in a new environment.

C. Comparison of sampling strategies

Figure 15 and Table VII use the same 130 configurations as Table VI. Median and P90 summarize the configuration-wise RMSE values; ϵ_{90} summarizes their pooled absolute trial errors.

Figure 15 and Table VII compare the two strategies across all configurations, retaining cases where systematic sampling does not improve. At 5% bandwidth and full span, median RMSE falls from 3.36° for pooled sampling to 2.37° for systematic sampling. The median paired reduction is 17.5%, with 128/130 configurations improving. This is the median of the configuration-wise percentage reductions, rather than the percentage change between the two median RMSE values. The 90% absolute-error tolerance across the pooled trials decreases from 8.1° to 7.3°.

At the same observation budget, systematic sampling guarantees 50 distinct, approximately equally spaced positions, whereas pooled sampling permits repeated positions and uneven spatial gaps. This allocation improves the median RMSE in these records. The systematic method's P90 remains 7.22° at 5% bandwidth and full span, so some residual offsets remain. A lower median does not guarantee accuracy in every measurement.

These results compare two allocations of 50 observations against the same empirical WB benchmark; they do not establish absolute calibration accuracy or a guarantee for a new measurement environment.

VIII. CONCLUSION

This work connects dominant-path phase-error statistics with the design of spatial averaging for bandwidth-limited measurements. For a dominant direct or reflected component and a weaker aggregate residual whose relative phase is uniform and independent of amplitude, the derived distribution is symmetric with zero mean. In the indoor measurements targeting the LOS component, the normalized weak-perturbation approximation based on Rayleigh residual amplitudes yields smaller ECDF discrepancies than the fitted zero-mean Gaussian baseline across the evaluated pooled bandwidth–frequency configurations. Spatial averaging improves agreement with the paired full-bandwidth reference, including for circular estimates of relative array phase at a fixed observation count. At the same observation budget, systematic sampling at approximately equally spaced positions lowers median RMSE and the absolute-error tolerance required for a prescribed empirical coverage compared with pooled random sampling in these records. However, spatial correlation and finite coverage constrain the averaging benefit, and reduced variability does not ensure removal of residual bias. These findings support designing spatial coverage and observation allocation jointly for narrowband phase estimation and relative calibration. The calibration target includes any fixed geometric phase, and the empirical reference does not establish absolute hardware-phase accuracy. Validation with an independent phase reference and extension to larger arrays remain future work.

REFERENCES

[1] C. Qin, J. A. Zhang, X. Huang, K. Wu, and Y. J. Guo, "Fast angle-of-arrival estimation via virtual subarrays in analog antenna array," *IEEE Trans. Wireless Commun.*, vol. 19, no. 10, pp. 6425–6439, 2020.

[2] P. J. G. Teunissen, "The least-squares ambiguity decorrelation adjustment: a method for fast GPS integer ambiguity estimation," *J. Geod.*, vol. 70, no. 1–2, pp. 65–82, 1995. [Online]. Available: https://doi.org/10.1007/BF00863419

[3] P. de Jonge and C. Tiberius, "The LAMBDA method for integer ambiguity estimation: implementation aspects," Delft Geodetic Computing Centre, Delft Univ. Technol., Delft, The Netherlands, LGR-Series 12, Aug. 1996.

[4] L. Chen, X. Zhou, F. Chen, L.-L. Yang, and R. Chen, "Carrier phase ranging for indoor positioning with 5G NR signals," *IEEE Internet Things J.*, vol. 9, no. 13, pp. 10 908–10 919, 2022.

[5] A. Bedin, J. Widmer, M. Davila, M. Canil, and R. Ruiz, "Millimeter-scale absolute carrier phase-based localization in multi-band systems," 2025. [Online]. Available: https://arxiv.org/abs/2511.05204

[6] A. Bedin, J. Widmer, M. Davila, M. Canil, and R. Ruiz, "Millimeter-scale absolute carrier phase-based localization in multi-band systems," in *Proc. ACM/IEEE Int. Conf. Embedded Artif. Intell. Sens. Syst. (SenSys)*, 2026, pp. 229–243. [Online]. Available: https://doi.org/10.1145/3774906.3800494

[7] A. Fascista, B. J. B. Deutschmann, M. F. Keskin, T. Wilding, A. Coluccia, K. Witrisal, E. Leitinger, G. Seco-Granados, and H. Wymeersch, "Joint localization, synchronization and mapping via phase-coherent distributed arrays," *IEEE J. Sel. Topics Signal Process.*, vol. 19, no. 2, pp. 412–429, 2025.

[8] W. Li, J. Zhou, J. Xiong, Y. Xie, L. Wang, D. Zhang, and D. Zhang, "Rethinking WiFi-based angle estimation for robust passive indoor localization," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 9, no. 4, Dec. 2025. [Online]. Available: https://doi.org/10.1145/3770662

[9] A. Xhafa, F. Fabra, D. Egea-Roca, J. A. López-Salcedo, and G. Seco-Granados, "Experimental investigation of AoA estimation and antenna calibration with 5G NR signals using USRP devices," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–15, 2025, art. no. 5504515. [Online]. Available: https://doi.org/10.1109/TIM.2025.3569347

[10] B. Friedlander and A. J. Weiss, "Direction finding in the presence of mutual coupling," *IEEE Trans. Antennas Propag.*, vol. 39, no. 3, pp. 273–284, 1991. [Online]. Available: https://doi.org/10.1109/8.76322

[11] I. J. Gupta, J. R. Baxter, S. W. Ellingson, H.-G. Park, H. S. Oh, and M. G. Kyeong, "An experimental study of antenna array calibration," *IEEE Trans. Antennas Propag.*, vol. 51, no. 3, pp. 664–667, 2003. [Online]. Available: https://doi.org/10.1109/TAP.2003.809870

[12] H. Gao, W. Wang, W. Fan, F. Zhang, Z. Wang, Y. Wu, Y. Liu, and G. F. Pedersen, "Design and experimental validation of automated millimeter-wave phased array antenna-in-package (AiP) experimental platform," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–11, 2021. [Online]. Available: https://doi.org/10.1109/TIM.2020.3024428

[13] B. Yu, F. Liu, H. Yi, and Y. Shi, "Multi-tone amplitude-phase calibration for active phased array antennas in a compact multi-probe OTA chamber," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–8, 2025. [Online]. Available: https://doi.org/10.1109/TIM.2025.3577855

[14] R. Pöhlmann, S. Zhang, E. Staudinger, S. Caizzone, A. Dammann, and P. A. Hoeher, "Bayesian in-situ calibration of multiport antennas for DoA estimation: Theory and measurements," *IEEE Access*, vol. 10, pp. 37 967–37 983, 2022. [Online]. Available: https://doi.org/10.1109/ACCESS.2022.3164520

[15] M. Pan, S. Liu, P. Liu, W. Qi, Y. Huang, W. Zheng, Q. Wu, and M. Gardill, "In situ calibration of antenna arrays for positioning with 5G networks," *IEEE Trans. Microw. Theory Techn.*, vol. 71, no. 10, pp. 4600–4613, 2023. [Online]. Available: https://doi.org/10.1109/TMTT.2023.3256532

[16] M. Jokinen, O. Kursu, N. Tervo, A. Pärssinen, and M. E. Leinonen, "Over-the-air phase calibration methods for 5G mmW antenna array transceivers," *IEEE Access*, vol. 12, pp. 28 057–28 070, 2024.

[17] H. Dun, C. C. J. M. Tiberius, C. E. V. Diouf, and G. J. M. Janssen, "Design of sparse multiband signal for precise positioning with joint low-complexity time delay and carrier phase estimation," *IEEE Trans. Veh. Technol.*, vol. 70, no. 4, pp. 3552–3567, 2021.

[18] J. K. Ray, M. E. Cannon, and P. C. Fenton, "Mitigation of static carrier-phase multipath effects using multiple closely spaced antennas," *NAVIGATION*, vol. 46, no. 3, pp. 193–202, 1999. [Online]. Available: https://doi.org/10.1002/j.2161-4296.1999.tb02406.x

[19] T.-J. Shan, M. Wax, and T. Kailath, "On spatial smoothing for direction-of-arrival estimation of coherent signals," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 33, no. 4, pp. 806–811, 1985. [Online]. Available: https://doi.org/10.1109/TASSP.1985.1164649

[20] A. Bilich and K. M. Larson, "Mapping the GPS multipath environment using the signal-to-noise ratio (SNR)," *Radio Sci.*, vol. 42, no. 6, 2007, art. no. RS6003. [Online]. Available: https://doi.org/10.1029/2007RS003652

[21] R. H. Clarke, "A statistical theory of mobile-radio reception," *Bell Syst. Tech. J.*, vol. 47, no. 6, pp. 957–1000, 1968. [Online]. Available: https://doi.org/10.1002/j.1538-7305.1968.tb00069.x

[22] N.-F. Zhang, "Calculation of the uncertainty of the mean of autocorrelated measurements," *Metrologia*, vol. 43, no. 4, pp. S276–S281, 2006. [Online]. Available: https://doi.org/10.1088/0026-1394/43/4/S15

[23] T. Schmidl and D. Cox, "Robust frequency and timing synchronization for ofdm," *IEEE Trans. Commun.*, vol. 45, no. 12, pp. 1613–1621, 1997.

[24] J.-J. van de Beek, M. Sandell, and P. O. Börjesson, "ML estimation of time and frequency offset in OFDM systems," *IEEE Trans. Signal Process.*, vol. 45, no. 7, pp. 1800–1805, 1997. [Online]. Available: https://doi.org/10.1109/78.599949

[25] P. Berens, "CircStat: A MATLAB toolbox for circular statistics," *J. Stat. Softw.*, vol. 31, no. 10, pp. 1–21, 2009. [Online]. Available: https://doi.org/10.18637/jss.v031.i10